

Assignment - 2

1. Given an integer array nums, sort it such that:

1. Elements with **lower frequency come first**
2. If frequencies are equal, sort by **higher numerical value first**

Constraints

- $1 \leq \text{nums.length} \leq 10^5$
- $-10^4 \leq \text{nums}[i] \leq 10^4$

2. Given an array, count how many pairs (i, j) exist such that:

$i < j$ AND $\text{arr}[i] > \text{arr}[j]$

Constraints

- $n \leq 10^5$

3. An array is k-sorted if every element is at most k positions away from its sorted position. Sort the array efficiently.

Constraints

- $n \leq 10^6$
- $k \ll n$

4. Given arrays arr1 and arr2, sort arr1 such that:

- Elements present in arr2 appear first
- They follow the same order as in arr2
- Remaining elements appear at the end in ascending order

5. Given a string, sort characters in decreasing order of frequency.

Example

Input: "tree"

Output: "eert" or "eetr"

6. You are given a file containing **billions of integers** that cannot fit in memory.

How do you sort it?

7. Given an array of integers arr of size n, write a C program to **stably sort** the array such that all **negative numbers appear before positive numbers**, while **preserving the relative order** of both negative and positive elements.

Input

- Integer n
- Array arr of size n

Output

- The rearranged array

8. Given an array of n integers, write a C program to determine whether the array can be sorted in **non-decreasing order** by performing **exactly one subarray reversal**.

Input

- Integer n
- Array arr of size n

Output

- Print YES if possible, otherwise print NO

9. Given an array of integers, write a C program to **sort the array according to the absolute difference from a given value k**.

If two numbers have the same absolute difference, the **larger number must come first**.

Input

- Integer n

- Array arr of size n
- Integer k

Output

- The sorted array

10. You are given two sorted arrays A and B of sizes n and m respectively.

Write a C program to **merge them into a single sorted array without using any extra space.**

Input

- Integers n, m
- Sorted array A
- Sorted array B

Output

- Arrays A and B should be sorted such that all elements of A are smaller than or equal to elements of B